QUANDELA × EPFL

Our Challenge FOR YOU – Details

EPFL Hackathon 2026

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Become a Quandela Ambassador
& Lead the Next Quantum Generation

QUANDELA KNIGHTS

The Knights

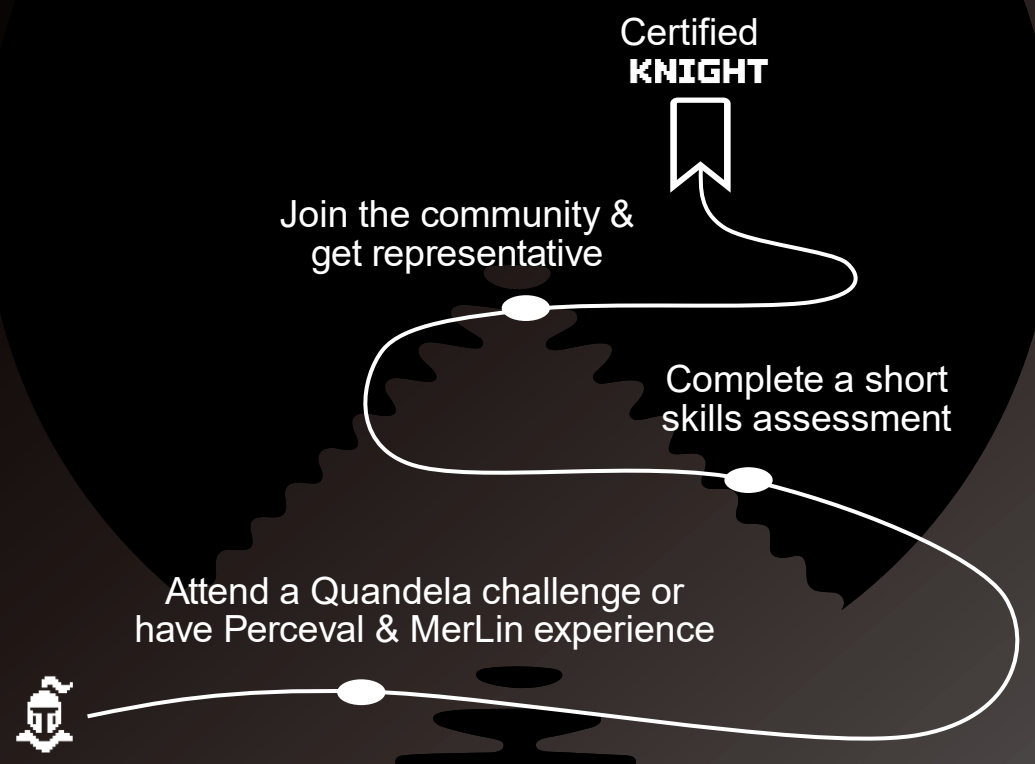
- Certified Quandela community
- Trusted to mentor, supervise & represent
- Local ambassadors

Missions

- Mentor teams
- Run workshops
- Supervise student projects
- Provide feedback
- Help building local community hubs

Incentives

- Paid role for selected events
- Exclusive trainings
- Become part of a global network
- Shape photonic quantum computing



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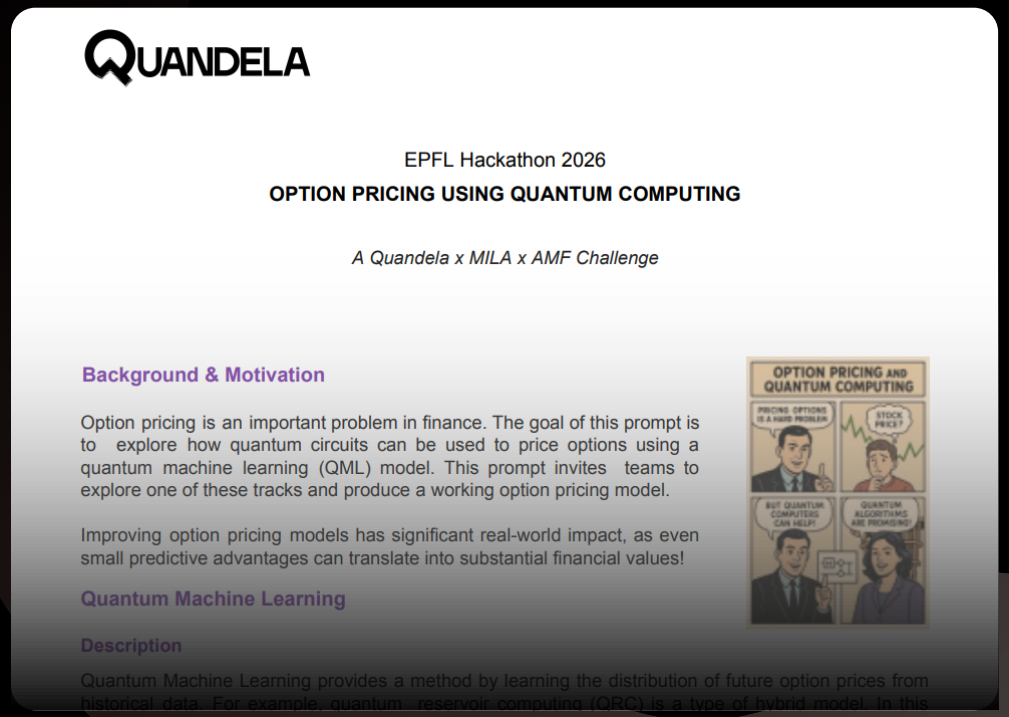
Swaptions Challenge
Option Pricing with Quantum Machine Learning

You will receive the slides afterwards – all information will be provided as a PDF together with the dataset and other materials!



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EPFL Hackathon 2026

OPTION PRICING USING QUANTUM COMPUTING

A Quandela x MILA x AMF Challenge

Background & Motivation

Option pricing is an important problem in finance. The goal of this prompt is to explore how quantum circuits can be used to price options using a quantum machine learning (QML) model. This prompt invites teams to explore one of these tracks and produce a working option pricing model.

Improving option pricing models has significant real-world impact, as even small predictive advantages can translate into substantial financial values!

Quantum Machine Learning

Description

Quantum Machine Learning provides a method by learning the distribution of future option prices from historical data. For example, quantum reservoir computing (QRC) is a type of hybrid model. In this

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Swaptions Challenge
Option Pricing with Quantum Machine Learning

In finance, a **tiny improvement** in forecasting accuracy **can be worth millions**. What if quantum machine learning could provide that edge?

Option pricing is not an academic toy problem, but central to:

- Risk management
- Derivatives trading
- Portfolio hedging
- Interest-rate markets

And yet forecasting remains extremely difficult!

This challenge asks: Can quantum models capture structure that classical ones struggle with?

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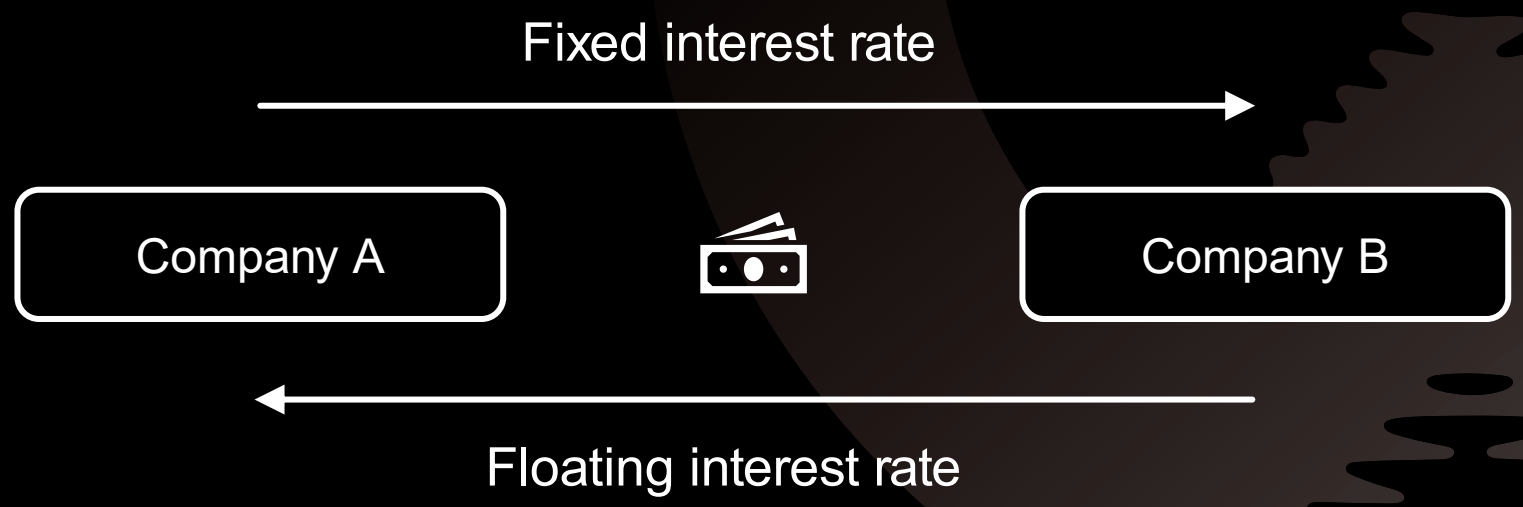
Swaptions Challenge
Option Pricing with Quantum Machine Learning

- An interest rate swap is a contract for which one party will pay a fixed interest rate, and the other will pay a floating interest rate.
- An option is the right (but not the obligation) to do something in the future.



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Swaptions Challenge
Option Pricing with Quantum Machine Learning

- A synthetic dataset of swaptions prices was generated with a realistic model and will be provided to the teams.

Goals

1. Predict future swaption prices!
2. Predict the values of missing data in the dataset!

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Swaptions Challenge
Option Pricing with Quantum Machine Learning

- The options on interest rate swaps (i.e. swaptions) have a tenor and a maturity
- The maturity is the moment at which an option can be exercised
- The tenor is the length of time of the interest rate swap payments
- MAIN TASK: PREDICT FUTURE DATA, predicting missing data is optional

Tenor : 10; Maturity : 30	Tenor : 15; Maturity : 30	Tenor : 20; Maturity : 30	Tenor : 25; Maturity : 30	Tenor : 30; Maturity : 30	Date
0,32239296	0,34585919	0,35916232	0,34667048	0,33767007	01-01-2050
0,3281438	Missing data !!	0,36519689	0,35099282	0,3408224	01-02-2050
0,32543597	0,34891453	0,36223568	Missing data !!	0,33902681	01-03-2050
0,32977101	0,3532459	0,36610005	0,3514044	0,34046496	01-05-2050
Future data	Future data	Future data	Future data	Future data	01-06-2050
Future data	Future data	Future data	Future data	Future data	01-08-2050

Use quantum machine learning techniques such as quantum reservoir or similar

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Swaptions Challenge
Option Pricing with Quantum Machine Learning

- Encode input window (e. g., past 20 trading days of features) into an optical/quantum reservoir
- Collect reservoir states (e. g., photon-count histograms / mode quadratures / qubit expectation values)
- Train linear readout (ridge/Lasso) to :
  (a) predict 1-10 day horizon prices; you may use walk-forward and compare to gradient boosted trees and small LSTMs
  (b) impute gaps
- Report accuracy and memory capacity vs. reservoir parameters (depth, nonlinearity, noise)

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Swaptions Challenge
Files to Submit

Excel template filled with results

Python Code & Presentation

Please upload a folder with the name of your team and which contains all files!

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Swaptions Challenge
Judging Criteria

Technical Aspects

- How accurate is the model?
- How complex is the quantum algorithm?
- Can the architecture serve users at reasonable scale?
- Etc.

Originality & Uniqueness

- How unique is this project compared to others?
- Did the team attempt something new or difficult?

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Swaptions Challenge
Judging Criteria

Usefulness & Complexity

- How useful is the project and how well-designed is it?
- How functional is it at the time of judging?

Presentation

- How well did the team present their project?
- Where they able to explain their decisions?

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Swaptions Challenge
Judging Criteria

Scan for access to the challenge folder with all the resources:

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